



Instore Payment Integration APIs

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Document Control

Version	Date	Author	Change Summary
1.0	May 30, 2023	Seun Matt	First Release

Document Approval

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Table Of Acronyms

Abbreviation	Definition
POS	Point Of Sale
FSP	Financial Service Provide
MSISDN	Mobile Station Integrated Services Digital Network

MoMo	Mobile Money
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1. Introduction

The purpose of this document is to provide a guide to Third Party System providers to allow them to interface with the Cellulant Instore Platform. It will be used to ensure that the Third Party System providers understand the standards that the interface adheres to, as well as explain the mechanism for connection that is recommended to allow the interface to be implemented.

A merchant integrates by consuming API resources for initiating a payment and configuring a callback URL to which all its Instore payments will be sent for acknowledgement.

2. Architecture

The Tingg Instore APIs are RESTful and utilise application/json content type in both the request and response body. Every request should have the Authorization header with a valid access token.

None

Authorization: Bearer accessToken

The only exception is the OAuth endpoint to generate the access token. Please note that the URLs in this document are for Tingg's sandbox environment. It is stable and will perform an actual payment - hence testing with dummy data is not supported currently.

Once the integration is complete on the sandbox environment, and UAT has been passed. Tingg will provide you with production credentials and endpoints.

2.1 Initiating USSD Push Payments from a POS Device

1. Merchants receiving the goods and services will enter the customer's mobile number, amount and any other reference. The POS device would have been pre-configured with a store code, store name and unique terminal ID
2. Given the aforementioned information, the merchant then initiates a payment request from the POS, that is sent via a secure HTTPS connection to the Tingga platform to charge customers. Tingga immediately responds with the status pending
3. Tingga platform then routes the request to the appropriate Financial Service Provider (FSP), in this case, the Mobile Money (MoMo) platform
4. The FSP then initiates a funds authorization USSD prompt to the customer which will prompt them to provide their personal PIN to approve the debit of their MoMo wallet for the billed amount
5. Once approved, the FPS will send the results via callback to the Tingga platform with information about the status of the transaction
6. Tingga will then process and send an Instant Payment Notification (IPN) to the merchant's POS for immediate confirmation and fulfilment of the transaction
7. The merchant can then print a receipt of the transaction and issue it to the customer
8. Tingga also sends a confirmation SMS to the customer
9. Optionally Tingga can send a similar notification to pre-configured merchants' mobile numbers

3. API Specification

The process of initiating a USSD payment transaction involves calling the following API endpoints sequentially:

HTTP Method	API Route	Description
POST	https://accounts.sandbox.tingga.africa/api/v1/oauth/token	This endpoint will get the OAuth access token for use in subsequent calls

GET	https://payments-instore.sandbox.tingg.africa/proxyPaymentMethods	API to get available MNOs the merchant can initiate Payments for
POST	https://payments-instore.sandbox.tingg.africa/initiateMerchantPayment	API to initiate a payment transaction
POST	To be provided by the merchant	This is the call

3.1 Authentication

The Tingga platform authentication is based on the OAuth2.0 protocol. The merchant will be provided with a set of credentials for generating valid access and refresh tokens they can use for subsequent API calls.

All subsequent calls should have the Authorization header with the value of the returned access token

URL:

https://accounts.sandbox.tingg.africa/api/v1/oauth/token?grant_type=password&client_id=payments&username=gatheca@cellulant.com&password=gatheca@cellulant.com&scope=read

HTTP Method: POST

The API expects the following parameters. The parameters encoded as a form post.

Parameter	Type	Description
Grant Type { grant_type }	String	The expected grant type are password when using credentials, authorization_code when using provided code, refresh_token when using a refresh token. client_credentials for app authentication
Client ID { client_id }	String	The Tingga Product Code requesting for access. When using other grant types. But using client_credentials , then it is the api user username

Client Secret { client_secret }	String	Required only when using client_credentials grant type. Refers to api user password.
Authorization Code { code }	String	The authorization code. Required when grant type is authorization_code .
Redirect URL { redirect_uri }	String	The original redirect link on request. Required when grant type is authorization_code .
Username { username }	String	The username for the current user. Required when grant type is password .
Password { password }	String	The password for the current user. Required when grant type is password .
Refresh Token { refresh_token }	String	The refresh token that was issued. Required when grant type is refresh_token .
Scope { scope }	String	The scope for the token. Not required.

POST /oauth/token

grant_type=password&client_id=engagement&username=user&password=password&scope=

Here is a sample call in staging

```
curl -X POST
'https://accounts.sandbox.tingg.africa/api/v1/oauth/token?grant_type=password&client_id=payments&username=gatheca@cellulant.com&password=gatheca@cellulant.com&scope=instore'
```

Response Body

Parameter	Type	Description
token_type	String	The value will always be “bearer”
expires_in	long	Time the tokens will expire in seconds
access_token	String	The access token for subsequent requests
refresh_token	String	The OAuth refresh token for generating new access token

error	String	This will be present only when the request is not successful and it will contain a short error code
error_description	String	This will be present only when the request is not successful and it contains the description of the error

Sample Request/Response

None

Request

```
{
  "username": "instore-demo-ke",
  "password": "g,2UWrVa!wK%`s!8U5a4W9#!0V-CN^3.",
  "grant_type": "password"
}
```

Successful Response

```
{
  "token_type": "bearer",
  "expires_in": 3600,
  "access_token":
    "eyJraWQiOiJjYXMtcGsiLCJhbGciOiJSUzUxMiJ9.eyJzdWIiOiI5NTQ3NSIsImF1ZCI6ImlvbiLCJsZXZlbnkiOiIkdVUURPTUVSIiwiaXNzIjoiaHR0cDovL2Nhcy1hcGkuYXBwcm92YWw6ODA4MC8iLCJuYW1lIjoiaW5zdG9yZS1kZWVtLWtlIiwidXR5cGUiOiJTQUU5EQk9YXOFQSStmNsaVVudC16IklOU1RPUkVERU1PS0UiLCJleHAiOjE2ODU0MjQwMjQzMjQ1MTY4NTQyMDQyNCwiY3VzdF9pZCI6MTI4ODUsImp0aSI6ImRjYWI2ZjFmLWMYdGltNDAYMilicOWQyLTdhYzRiZDM0YjllNzgZMTQ0NDg1LTRmOWUtNDI1YS1hOTRjLTA1NDc1YjFkNDNiMCJ9.vE_vL_rvMX9FMDHY-jLW6YzspuTrEZ19Bt3Chbsot_X1QaTuE42yMAgcKJOQ-z45fvLZA123dnIBHg0OoYn7dfCSmlyK2JJDEcQ4mypb-wB7E3gGeWgp6R_Zx6MnFWWY7Kys7qU6G1qVGRTla8-oBM8CJ8K92K0ZbFvczkv_UiwkBN0z6DQHvkFYetWJR16gPouZq6IEU9hixzdLNNr6Vp0W3En_qgkdlY7XWnLWA7A1JPxtA8oeBEhpSQkLvWzP2jt6EKDN8nP3kgGT2Qm-qFhPeSWnnuHMICTD0BvPe1SrS_ooLd4zLUGcPFsgdTAAxv4LmxS_BGOhkHn6pWAXw",
  "refresh_token":
    "OEVRbkM5bDM0Q0IVUDEvcTR2dkY3QU5PV2ZGdVVJVVGQxRDBGNmQ0WjlDaWF2emMOY1pocGI4NU8wSXvtT2pSbtJNM3QzUXlwUEJQQnJUSTIzejM4UEE9PQ=="
}
```

Failed Response


```
{
  "error": "invalid_request",
  "error_description": "Invalid credentials"
}
```

3.2 Payment Methods

Tingg platform supports a plethora of mobile money operators (MoMo) and wallets. It is very important as a client of our API to know which MoMo is configured for your account. Each MoMo is referred to as a Payer client on Tingg and they each have an assigned code for initiating a payment. Not using the right payer client code will lead to failed payments as customers will not get the authorisation prompt from their respective MNO.

URL: <https://payments-instore.sandbox.tingg.africa/proxyPaymentMethods>

HTTP Method: GET

Content-Type: application/json

Request Parameters

Parameter	Type	Required	Description
countryCode	varchar	Yes	This is the merchant's 3 character country code e.g. KEN for Kenya

Response Body

Parameter	Type	Description
data.*.customer.customerID	varchar	This is the internal ID assigned to this payer client
data.*.customer.customerName	varchar	The full name of the payer client
data.*.customer.customerLogo	varchar	The URL for the customer's logo

data*.customer.screenName	varchar	Name suitable for showing the end-customer
data*.customer.customerCode	varchar	The payer client's code. This data is to be used when initiating a Payment. Please pay attention to this one as it is required as an input when initiating a customer's debit
data*.currency.*	object	This object holds the currency information of the payer client
data*.countryDTO.*	object	Information about the payer client's country

Note: Other response data points not described in the aforementioned table but returned as part of the API response are for Tingg's internal processes and can be safely ignored unless otherwise stated.

Sample Request/Response

None

GET

<https://payments-instore.sandbox.tingg.africa/proxyPaymentMethods?countryCode=KEN>

Response

```
{
  "data": [
    {
      "paymentMethodID": 5252,
      "acknowledgePayer": false,
      "type": {
        "paymentMethodTypeID": 1,
        "paymentMethodName": "Bank",
        "paymentMethodTypeCode": "BANK"
      },
      "customer": {
        "customerID": 11038,
```

```

    "customerName": "Safaricom Kenya Limited",
    "customerCode": "SAFKE",
    "customerLogo":
    "https://customers.sandbox.tingg.africa/api/v1/app/images/2b79ef81-f951-43cb-98
    65-a3be705a494a",
    "mapping": {
      "id": 7,
      "code": "SAFKE"
    },
    "onboardingLevel": {
      "onboardingLevelID": 7,
      "onboardingLevelCode": "live",
      "onboardingLevelName": "Live"
    },
    "screenName": "Mpesa Checkout"
  },
  "currency": {
    "currencyID": 70,
    "currencyName": "Kenyan Shilling ",
    "currencyCode": "KES"
  },
  "countryDTO": {
    "countryID": 117,
    "countryName": "Kenya",
    "countryCode": "KEN",
    "countryAbbv": "KE"
  },
  "service": null,
  "serviceLimit": null
}
],
"message": "Service payment methods fetched successfully",
"success": true,
"statusCode": 200
}

```

Failed Response

```
{
```

```
"success": false,  
"statusCode": 172,  
"message": "countryCode is required",  
"data": null  
}
```

3.3 Counter Details

This endpoint will provide information about the merchant, store, counter or till the customer wants to pay. The information returned from the API should be displayed to the customer to help them confirm they're paying for the right service.

URL:

<https://instore-management.sandbox.tingq.africa/v2/counterCodes/{counterCode}/compact>

HTTP Method: GET

Content-Type: application/json

Path Parameter

Parameter	Type	Required	Description
counterCode	varchar	Yes	This is the unique identifier for the Terminal or Counter the customer wants to make payments into

Request Headers

Parameter	Type	Required	Description
X-Country-Code	varchar	Yes	This is the merchant's 3-character country code e.g. KEN for Kenya

Response Body

Parameter	Type	Description
data.storeName	varchar	This is the name of the store the counter code belongs to as configured on the Tingga web portal during creation
data.storeCode	Long	The unique Tingga identifier for the Store where the Counter belongs
data.counterName	varchar	The name of the Counter as configured on the Tingga web portal during creation
data.counterCode	varchar	The unique identifier for the counter. The same as provided in the URL path parameter
data.countryCode	varchar	The country where the counter is located e.g. KEN
data.isReference	Long	The a flag to indicate if the counter take in a reffene or not
data.address	String	The physical address associated with the Counter as configured on the Tingga web portal

Sample Request/Response

None

Request

GET

<https://instore-management.sandbox.tingga.africa/v2/counterCodes/{counterCode}/compact>

Successful Response

```
{
  "success": true,
  "statusCode": 200,
  "message": "Success",
  "data": {
    "storeName": "Test Store ",
    "counterName": "Counter",
    "counterCode": "1001",
    "address": "test ",
    "storeCode": 1234,
```

```
"isReference": false,
"countryCode": "KEN",
"alias": "MOMO",
"serialNumber": null,
"referenceName": null,
"isTicketing": false,
"isProduct": true,
"product": [ {
  "counterCodeOptionID": 198,
  "price": 0.0,
  "expireDate": null,
  "counterCodeDetailID": 678,
  "optionName": "Petrol"
}],
"ticketing": []
}
}
```

Failed Response

```
{
  "success": false,
  "statusCode": 404,
  "message": "Counter code not found",
  "data": null
}
```

3.4 Reference Validation

This is the endpoint that will be used to validate raw user input based on external merchant records. The merchant will implement an API on their end that will do the validation based on the request sent from Cellulant and return the specified response body below. The API is required to conform to the format shown.

The API endpoint will be configured on our end, along with any custom headers, e.g an API key

URL: Merchant provided integration to their own API

HTTP Method: POST

Content-Type: application/json

Request Body

Parameter	Type	Required	Description
userProvidedValue	String	Yes	The raw input provided by the user (e.g., student ID, admission number).
countryCode	String	Yes	ISO country code identifying the country context (e.g., "KEN").
clientId	String	Yes	Unique identifier for the client/school making the request.
extraData	Object	No	Additional data for validation.
scenarioType	String	No	The validation context type, e.g, COUNTER_CODE
scenarioValue	String	No	Contextual value for the scenario (e.g., "789").

Request Headers

Parameter	Type	Required	Description
Authorization	String	No	API Key (or an equivalent shared key)
Content-Type	String	Yes	application/json

Response Body

Parameter	Type	Required	Description
success	Boolean	Yes	This will be true if the request is successful or false if it is not successful

message	String	Yes	A description of the API response. It will contain an error message if the request is not successful
errors	Array of Strings	Yes	List of error messages if validation failed.
fields	Object Map<String, String>	Yes	Key-value pairs of validated fields (e.g., studentId, studentName).

Sample Request/Response

JSON

Request Body

```
{
  "userProvidedValue": "STU-2025-001",
  "countryCode": "KE",
  "clientId": "12345",
  "extraData": {},
  "scenarioType": "COUNTER_CODE",
  "scenarioValue": "789"
}
```

Successful Response

```
{
  "success": true,
  "message": "Student record validated successfully",
  "errors": [],
  "fields": {
    "studentId": "STU-2025-001",
    "studentName": "John Doe",
    "gradeLevel": "Form 2",
    "feeAmount": "ZMW 2000"
  }
}
```



```
Failed Response
{
  "success": false,
  "message": "Invalid student data",
  "errors": [
    "Grade level must be between Form 1 and Form 4"
  ],
  "fields": {}
}
```

All failure responses should return a sensible HTTP failure status code (e.g, 400) with indicated validation errors and success set as false.

3.5 Initiate Payment

This is the endpoint that will raise a debit request and send a prompt to a customer's phone to authorize the transaction with their private MoMo wallet PIN.

URL: <https://payments-instore.sandbox.tingg.africa/initiateMerchantPayment>

HTTP Method: POST

Content-Type: application/json

Request Body

Parameter	Type	Required	Description
counterCode	Long	Yes	This is the code (Terminal ID) that has been assigned to the POS device
msisdn	String	Yes	This is the phone number of the customer
amount	Double	Yes	This is the amount to be paid by the customer to the merchant. The amount must be greater than or equal to 1
payerClientCode	String	Yes	This is the payment method's <i>data.customer.customerCode</i> that the customer has selected to make a payment with e.g TINGGMTN or SAFKE

reference	String	No	This is an optional system reference you can provide when initiating a transaction.
------------------	--------	----	---

Request Headers

Parameter	Type	Required	Description
X-Country-Code	String	Yes	This is the 3-char ISO country code e.g KEN for Kenya
Currency-Code	String	Yes	This is the 3-char ISO currency code i.e. ZMW
Request-Origin	String	Yes	This is the request origin ID of the request. Please use TINGG_INSTORE_INTEGRATION during development. You will be provided with another value for production

Response Body

Parameter	Type	Type	Description
success		Boolean	This will be true if the request is successful or false if otherwise
statusCode		Integer	The HTTP status code of the request
message		String	A description of the API response. It will contain an error message if the request is not successful
data.merchantTransaction ID		Long	An ID generated from Tingg for this payment request

Sample Request/Response

None

Request

```
{
  "msisdn": "254705568254",
  "amount": 1,
  "counterCode": "12150",
  "payerClientCode": "SAFKE",
  "reference": null
}
```

Successful Response

```
{
  "success": true,
  "statusCode": 200,
  "message": "A payment request of KES 1.0 has been sent to
254705568254. Ask the customer to authorize the payment on their
phone.",
  "data": {
    "merchantTransactionID": 1556930
  }
}
```

Failed Response

```
{
  "success": false,
  "statusCode": 400,
  "message": "We are unable to process your request at this time,
please try again later.",
  "data": "Required request header 'X-Country-Code' for method
parameter type String is not present"
}
```

3.6 Instant Payment Notification (IPN)

Following a successful payment initiation and authorization by the customer, Tingga will receive a notification from the customer's Financial Service Provider (FSP). Tingga will process this transaction and can, in turn, send a callback request to the merchant's system.

The merchant will have to configure their system URL for receiving IPN from Tingga. Please take note of the response expected from the merchant's callback. In cases where the merchant's system is not able to process the IPN, Tingga will automatically retry sending the notification again with a backoff policy.

URL: To be configured by the merchant

HTTP Method: POST

Content-Type: application/json

IPN Request Body

Parameter	Type	Required	Description
merchantName	String	No	Name of the merchant, eg. Shoprite
storeCode	Long	Yes	This code identifies the Store on Instore Tingga6, it is unique
storeName	String	Yes	This is the name given to the Store when creating it on the Tingga portal. It can be any name chosen by the merchant
counterCode	String	Yes	This is the counterCode provided when initiating the transaction. It identifies a counter or Till in the Store
amount	Double	Yes	The amount of money paid by the customer towards the goods/service being provided
currencyCode	String	No	The 3-char ISO currency code for the transaction e.g ZMW, USD, GHS
reference	String	No	This is the same reference passed

			during payment initiation. It is propagated to the merchant's system for reconciliation
beepTransactionID	String	No	Tingg transaction ID generated to track payments
payerTransactionID	String	No	Wallet-based transactionID which is sent from the payer's Mobile money wallet
merchantNotificationID	Long	Yes	This is the unique number from Tingg that identifies this IPN request
msisdn	String	No	This is the phone number the customer used to authorise a MoMo payment. It will be available for MoMo payments
narration	String	No	The payment's narration
extraData	Object	No	This holds any metadata passed during PostPaymanet
counterName	String	No	This holds the name of the counter if it exists
statusDescription	String	No	Provides information regarding the transaction that has been done
merchantTransactionID	String	No	This is a merchant UNIQUE identifier that can be used by the initiator of the payment for reconciliation (This Field is Configurable, it can be added or removed from the response body)
paymentDate	String	No	This holds the time when the payment was logged on our end (This Field is Configurable, it can be added or removed from the response body)
statusCode	Integer	Yes	This code identifies the status of the request

IPN Response Body

Parameter	Type	Required	Description
statusCode	Integer	No	This code identifies the status of the request

merchantTransactionID	String	Yes	This should be the same data.merchantTransactionID in the response of the call to initiate a payment in 3.4
statusDescription	String	Yes	This is the description of the status
callbackResponseID	String	Yes	This should be the same value as the merchantNotificationID in the callback request

IPN Response Status Code

Status Code	Status Description
188	Successfully acknowledged
189	Failed acknowledgement

Sample Request/Response

None

Request

```
{
  "merchantName": "Shoprite",
  "storeCode": 75,
  "storeName": "Store Name",
  "counterCode": 19943,
  "counterName": "Name of Counter",
  "amount": 500,
  "currencyCode": "ZMW",
  "reference": "reference-123",
```

```
"beepTransactionID": "121313",
"payerTransactionID": "REU2QZKDNK",
"statusCode": 140,
"statusDescription": "successfully processed",
"msisdn": "x",
"narration": "Customer has paid ZMW 500 to your store",
"merchantNotificationID": 112,
"merchantTransactionID": "89765",
"paymentDate": "2025-01-13 12:55:00.0",
"extraData": {
  "reference": "ecf1821",
  "productCode": "Tuition",
  "counterCode": "15216",
  "storeCode": "6383"
}
}
```

Response

```
{
  "statusCode": "188",
  "merchantTransactionID": "90000013",
  "statusDescription": "transaction acknowledged successfully",
  "callbackResponseID": "1323"
}
```

3.7 Payment History

This API is meant to be called in situations where the IPN did not come as expected due to errors or internal incidents. **The API is not to be used as the primary means of confirming transactions.**

Please note the following:

- By default, the API will return 1000 transactions conducted in the last 24 hours
- The difference between the **fromDate** and the **toDate** can not exceed seven (7) days
- The pagination is controlled by you as we will not be returning the **total** parameter to you due to technical reasons. How do I know that's all the records? When the combination of your page and size parameter returned an empty array.

URL: <https://payments-instore.sandbox.tingg.africa/merchants/payments>

HTTP Method: GET

Content-Type: application/json

Request Headers

Parameter	Type	Required	Description
X-Country-Code	String	Yes	The three-letter country code where the transaction is done. Typically country of the business e.g. KEN, NGA, ZMB, TZA, UGA

Request Param

Parameter	Type	Required	Description
fromDate	String	No	This is the start date for which transactions are to be returned. The expected format is yyyy-MM-dd HH:mm:ss for example 2023-11-08 00:00:50
toDate	String	No	This is the end date for which transactions are to be returned. This is required if fromDate is provided. The expected format is yyyy-MM-dd HH:mm:ss for example 2023-11-08 00:00:50

payerTransactionID	String	No	Filter for transactions belonging to the provided payerTransactionID
transactionID	String	No	Filter for transactions belonging to the provided transactionID
counterCode	String	No	Filter for transactions belonging to the provided counterCode
storeCode	String	No	Filter for transactions belonging to the provided storeCode
page	int	No	The page to view starting from 1
size	int	No	The size of the result to return. Maximum of 1000
mobileNumber	String	No	Add this request param to return transactions done by this mobile number e.g. 260779393002. This field is called msisdn in the returned response

Response Param

Parameter	Type	Description
data.*.transactionID	String	This is the transactionID of the transaction
data.*.payerTransactionID	String	This is the transactionID of the transaction
data.*.currencyCode	String	This is the currencyCode of the transaction
data.*.statusDescription	String	This is the statusDescription of the transaction
data.*.requestOriginName	String	This is the requestOriginName of the transaction
data.*.amountPaid	Double	This is the amount paid for the transaction
data.*.paymentStatus	Integer	This is the paymentStatus of the transaction
data.*.payerNarration	String	This is the requestOriginName of the transaction

data.*.paymentDate	String	The payment date connected with the transaction. E.g. 2023-11-20T10:06:41
data.*.storeCode	String	This is the identifier of the Store connected to this transaction
data.*.counterCode	String	This is the identifier of the Counter connected to this transaction
data.*.msisdn	String	This is the mobile number that performed the transaction for Mobile Money payment methods
data.*.merchantTransactionID	String	This is the same ID returned from initiating a Mobile Money transaction described in section 3.4
data.*.requestOriginCode	String	This data point indicates the origin of the transaction. Common values are TINGG_INSTORE_USSD, TINGG_INSTORE_INTEGRATIONS
data.*.extraInformation	Object	This data point contains payment details metadata
data.*.paymentID	String	This data point indicated the paymentID that is unique per transaction
data.*.payerClientName	String	This data point indicates the payment option used
data.*.accountNumber	String	This data point indicates the account number paid for

Sample Request/Response

None

GET

<http://localhost:8080/merchants/payments?fromDate=2023-11-19%2000%3A00%3A50&toDate=2023-11-21%2004%3A55%3A50&payerTransactionID=RKK9UGBUM5&transactionID=1919772758&counterCode=12150&storeCode=542&page=1&size=1000>

--header 'X-Country-Code: KEN'

{

 "statusCode": 200,

 "message": "Successful",

```

"success": true,
"data": [
  {
    "transactionID": "1919772758",
    "payerTransactionID": "RKK9UGBUM5",
    "currencyCode": "KES",
    "statusDescription": "Payment Accepted",
    "requestOriginName": "Tingg Instore USSD",
    "amountPaid": 1.0,
    "paymentStatus": 140,
    "payerNarration": "The service request is processed successfully.",
    "paymentDate": "2023-11-20T10:06:41",
    "receiverClientCode": "INSTOREDEMOKE",
    "storeCode": "542",
    "counterCode": "12150",
    "msisdn": "254705568254",
    "merchantTransactionID": "4551274",
    "requestOriginCode": "TINGG_INSTORE_USSD",
    "extraInformation": {
      "reference": "Petrol",
      "productCode": "Petrol",
      "counterCode": "15683",
      "storeCode": "7351"
    },
    "paymentID": "120943554",
    "payerClientName": "Airtel Zambia",
    "accountNumber": "Petrol - Petrol"
  }
]

```

```

"transactionID": "100490924505320454",
  "payerTransactionID": "MP250707.1133.K54233",
  "currencyCode": "ZMW",
  "statusDescription": "Payment Accepted",
  "requestOriginName": "Tingg Instore USSD",
  "amountPaid": 3070.0,
  "payerNarration": "Payment of ZMW 3106.85 Till Number TINGZBM
Cellulant Zambia. Airtel Money bal is ZMW 10.86. TID : MP250707.1133.K54233.",
  "paymentDate": "2025-07-07T09:33:02",
  "receiverClientCode": "MOUNTMERUPETROLE_ZMB",
  "storeCode": "8670",
  "counterCode": "16339",
  "msisdn": "260974010106",

```

```
{
  "merchantTransactionID": "20031083",
  "requestOriginCode": "TINGG_INSTORE_USSD",
  "extraInformation": {
    "reference": "Petrol",
    "productCode": "Petrol",
    "counterCode": "15683",
    "storeCode": "7351"
  },
  "paymentID": "120943554",
  "payerClientName": "Airtel Zambia",
  "accountNumber": "Petrol - Petrol",
  "paymentStatus": 140
}
```

3.8 Single Payment Record

This endpoint is meant for transactions that are first initiated before they're completed/authorised by the end customer. For example, Mobile Money transactions.

You can call this endpoint with the ***data.merchantTransactionID*** received from initiating a transaction as described in section 3.4 above

URL:

<https://payments-instore.sandbox.tingg.africa/merchants/payments/{merchantTransactionID}>

HTTP Method: GET

Content-Type: application/json

Request Headers

Parameter	Type	Required	Description
X-Country-Code	String	Yes	The three-letter country code where the transaction is done. Typically country of

			the business e.g. KEN, NGA, ZMB, TZA, UGA
--	--	--	---

Response Param

Parameter	Type	Description
data.transactionID	String	This is the transactionID of the transaction
data.payerTransactionID	String	This is the transactionID of the transaction
data.currencyCode	String	This is the currencyCode of the transaction
data.statusDescription	String	This is the statusDescription of the transaction
data.requestOriginName	String	This is the requestOriginName of the transaction
data.amountPaid	Double	This is the amount paid for the transaction
data.paymentStatus	Integer	This is the paymentStatus of the transaction
data.payerNarration	String	This is the requestOriginName of the transaction
data.paymentDate	String	The payment date connected with the transaction. E.g. 2023-11-20T10:06:41
data.storeCode	String	This is the identifier of the Store connected to this transaction
data.counterCode	String	This is the identifier of the Counter connected to this transaction
data.msisdn	String	This is the mobile number that performed the transaction for Mobile Money payment methods
data.merchantTransactionID	String	This is the same ID returned from initiating a Mobile Money transaction described in section 3.4
data.requestOriginCode	String	This data point indicates the origin of the transaction. Common values are

		TINGG_INSTORE_USSD, TINGG_INSTORE_INTEGRATIONS
--	--	---

Sample Request/Response

None

GET <http://localhost:8080/merchants/payments/4551274>

--header 'X-Country-Code: KEN'

```
{
  "statusCode": 200,
  "message": "Operation successful",
  "success": true,
  "data": {
    "transactionID": "1919772758",
    "payerTransactionID": "RKK9UGBUM5",
    "currencyCode": "KES",
    "statusDescription": "Payment Accepted",
    "requestOriginName": "Tingg Instore USSD",
    "amountPaid": 1.0,
    "paymentStatus": 140,
    "payerNarration": "The service request is processed successfully.",
    "paymentDate": "2023-11-20T10:06:41",
    "receiverClientCode": "INSTOREDEMOKE",
    "storeCode": "542",
    "counterCode": "12150",
    "msisdn": "254705568254",
    "merchantTransactionID": "4551274",
    "requestOriginCode": "TINGG_INSTORE_USSD"
  }
}
```

Status Codes

Status Code	Status Message	Comment	Action
138	PaymentFailed	This indicates that the payment has failed	Failed

140	PaymentAccepted	This indicates that the payment was accepted	Successful
141	PaymentRejected	This indicates that the payment was rejected	Failed
219	PaymentEscalated	This indicates that the payment was escalated	Manual Recon
103	UnknownStatusCode	This indicated that the status returned is not configured with us.	Manual Recon
179	PendingReversal	This indicates the payment is pending a reversal	Manual Recon
216	TransactionManuallyFailed	This indicates the transaction was failed manually	Failed (Manual Recon)
217	TransactionManuallySuccess	This indicates that the payment was made successful manually	Successful (Manual Recon)
188	SuccessfullyDelivered	This indicates that the merchant received the payment successfully.	Manual Recon
189	FailedDelivered	This indicates that the merchant did not receive the payment successfully.	Manual Recon

3.1.2.3 PostPayment documentation

Endpoint-

<https://api-test.tingg.africa/v1/payments/postpayment>

Request Headers

Parameter	Type	Required	Description
X-Country-Code	String	Yes	Country code i.e. KEN,NGA,ZMB
Content-Type	String	Yes	application/json

Authorization	String	Yes	Bearer
----------------------	--------	-----	--------

Request Payload

Parameter	Type	Required	Description
MSISDN	Numeric	YES	It contains the customers mobile number
amount	Numeric	YES	It contains the amount being transacted
serviceID	Numeric	YES	This is the unique identifier tied to the service
narration	String	YES	It shows the narration sent
currencyCode	String	YES	This is a 3 char value representing the currency being used.
accountNumber	AlphaNumeric	YES	This contains the account number which is unique per service
originatorClientCode	String	NO	The client code for the client where the request originated from.
datePaymentReceived	DateTime	YES	It contains the date the payment was received
payerTransactionID	Numeric	YES	This is unique per MNO
customerNames	String	YES	The name for the customer making the payment.
isCombined	Boolean	NO	If the current payment is a combined payment.

paymentMode	String	YES	It shows the mode being used for payment
serviceCode	String	YES	unique code that identifies the service being paid for
clientCode	String	YES	the payer client code
requestOriginCode	String	YES	identifies the product a payment comes from eg TINGG_INSTORE_ZANACO_INTEGRATION from checkout
chargeBeneficiaries	Array	NO	holds data on the various commissions for the different beneficiaries of the charge amount
extraData	Object	NO	object with additional data that is used for service specific processing downstream
extraData.pushToOriginator			This value should be passed as true it is used to push the transaction to the merchant
extraData.storeCode			This code identifies the Store on Instore Tingg6, it is unique
extraData.counterCode			This is the counterCode provided when initiating the transaction. It identifies a counter or Till in the Store
extraData.requestOriginCode			This is a parameter that is sent alongside each notification to indicate the source of the transaction.
extraData.reference			This is a reference passed during payment initiation.

Sample Request

None

```
{
  "MSISDN": "256755548887",
  "amount": "120",
  "serviceID": 7566,
  "narration": "Paying for service",
  "currencyCode": "KES",
  "accountNumber": "<RRN>_<customerMSISDN>_<merchantCode>_PAYWITHZ",
  "chargeBeneficiaries": [],
  "originatorClientID": 7,
  "datePaymentReceived": "2023-09-22 08:19:19",
  "payerTransactionID": "{{${timestamp}}",
  "customerNames": "Liz Blinco",
  "isCombined": "0",
  "paymentMode": "MOBILE",
  "serviceCode": "citiugx",
  "clientCode": "MPEDDEV6419",
  "requestOriginCode": "TINGG_INSTORE_ZANACO_INTEGRATION",
  "extraData": {
    "pushToOriginator": true,
    "storeCode": 7225,
    "counterCode": 17688,
    "requestOriginCode": "TINGG_INSTORE_ZANACO_INTEGRATION",
    "reference": "Zanco Demo merchant store counter code 2024"
  }
}
```

Response Payload

Parameter	Type	Required	Description
statusCode	Integer	Yes	The api user authentication code
message	String	Yes	The api user authentication description
statusCode	Integer	Yes	Transaction processing status code

statusDescription	String	No	The merchant's receipt number
beepTransactionID	String	No	The transaction's unique identifier in CPG

Sample Response

None

```
{
  "message": "Authentication was successful",
  "success": true,
  "statusCode": 139,
  "data": {
    "statusDescription": "Payment pending acknowledgement",
    "payerTransactionID": "1702468912",
    "beepTransactionID": "522744971804041216"
  }
}
```

Status Codes

STATUS CODE	STATUS MESSAGE	COMMENT
144	Payment Received	This is a payment which has been posted to payments table successfully then pushed to post payment queue for further validation
132	Invalid Credentials. Authentication failed	Invalid Credentials. Authentication failed
132	Credentials payload section is missing or null.	This is for a payload which does not have any credentials sent with it.

	Authentication not possible.	
132	Username not supplied. Authentication not possible.	This is for a payload which was sent with the username missing on the credentials.
132	Password not supplied. Authentication not possible	This is for a payload which was sent with the password missing on the credentials.
132	Internal server error. Failed authentication.	This applies when we are not able to validate the user credentials on database due to database failure.
132	User with the provided username does not exist.	This applies when the username sent on the credentials does not exist.
131	Authentication was a success	Authentication was a success
139	Payment Pending acknowledgement	This is a payment which has been validated successfully by the post payment publisher
229	Duplicate payment found.	A payment with the same payer Transaction ID was found thus a duplicate payment
174	Error occurred while processing, contact Cellulant.	A generic error code
136	Payment ID not specified	This happens when the paymentID is not attached to the payload packet data.
111	Invalid amount specified	This happens when the invoice amount specified is invalid
109	MSISDN not specified	This happens when the packet data sent on the payload has no MSISDN
115	Currency code not specified	This happens when the currencyCode is not attached to the payload packet data.

110	Invalid MSISDN specified	This happens when one uses a mobile number with the wrong format
120	Account number not specified	This happens when the account number is not specified on the packet data on the payload
166	Service ID is not specified	This happens when the service ID is not present on the packet data on the payload.
415	Insufficient float balance	Float consumption failed due to insufficient float balance.

3.9 Request Origin Code

The request origin code depicts where the transaction originated from and it is optionally sent alongside the instant notification callback and payment history API response.

List of Request Origin Code

Request Origin Code	Description
TINGG_INSTORE_INTEGRATION	This code is for transactions that were initiated by merchant/partner systems via our secure

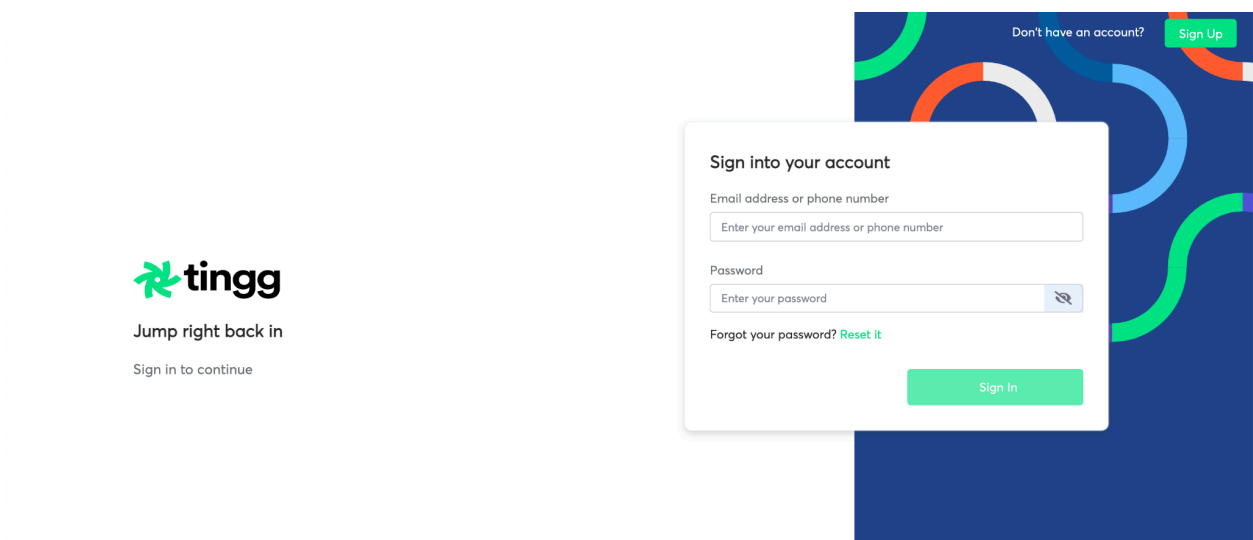
	API. If you're calling the initiate payment API in section 3.4, you'll most likely get this as the request origin code
TINGG_INSTORE_USSD	This request origin code is for transactions where the Customer dials a USSD code on their mobile phone to make a payment
TINGG_INSTORE_QRCODE	This is for transactions where a Customer scans a QRCode to make a payment
TINGG_INSTORE_ANDROID	This is for transactions initiated by Merchants or their Agents on our Mobile app
TINGG_INSTORE_WEB	This is for transactions initiated by Merchants or their Agents on our Web platform

None

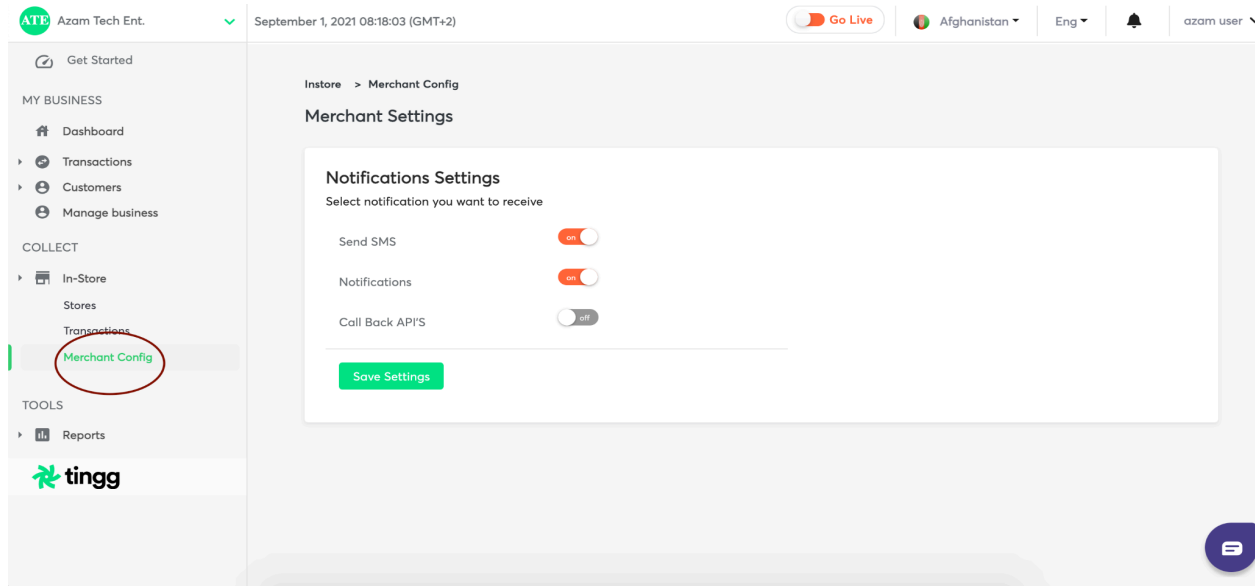
4. How to Configure IPN URL on the Tingga Web Portal

Please follow the steps below to configure the IPN endpoint/url on the Tingga web portal

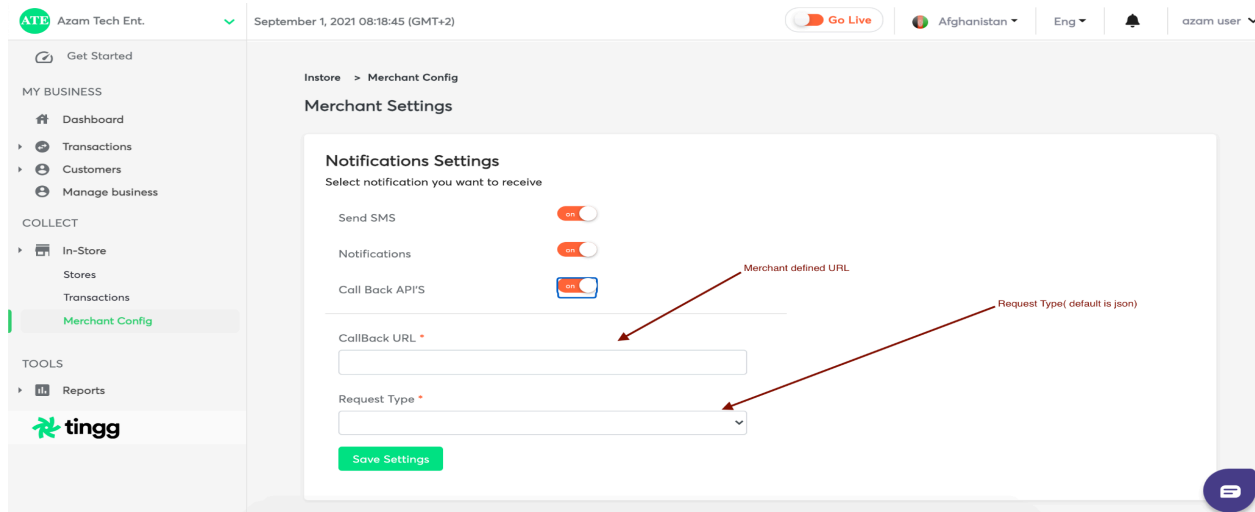
1. Login to Tingga through a web browser at <https://app.tingg.africa/>



- Navigate to the **Instore Menu** —> **Merchant Configurations** section to switch on the CallBack API toggle as shown below



- The next step is for the merchant to Configure the URL for receiving IPNs by completing the **Callback URL** field and selecting **JSON** as the protocol



By doing so the merchant will now be able to receive payment notification.

5. References

HTTP Method	API Production URL	Pre	API Route Production URL	Discription
POST	https://accounts.sandbox.tingg.africa/api/v1/oauth/token		https://accounts.tingg.africa/api/v1/oauth/token	This endpoint will get the OAuth access token for use in subsequent calls
GET	https://payments-instore.sandbox.tingg.africa/proxyPaymentMethods		https://payments.instore.tingg.africa/proxyPaymentMethods	API to get available MNOs the merchant can initiate Payments for
POST	https://payments-instore.sandbox.tingg.africa/initiateMerchantPayment		https://payments.instore.tingg.africa/initiateMerchantPayment	API to initiate a payment transaction
GET	https://payments-instore.sandbox.tingg.africa/merchants/payments		https://payments-instore.tingg.africa/merchants/payments	History API
GET	https://payments-instore.sandbox.tingg.africa/merchants/payments/{merchantTransactionID}		https://payments-instore.tingg.africa/merchants/payments/{merchantTransactionID}	Single Payment
POST	To be provided by the merchant		To be provided by the merchant	This is the call